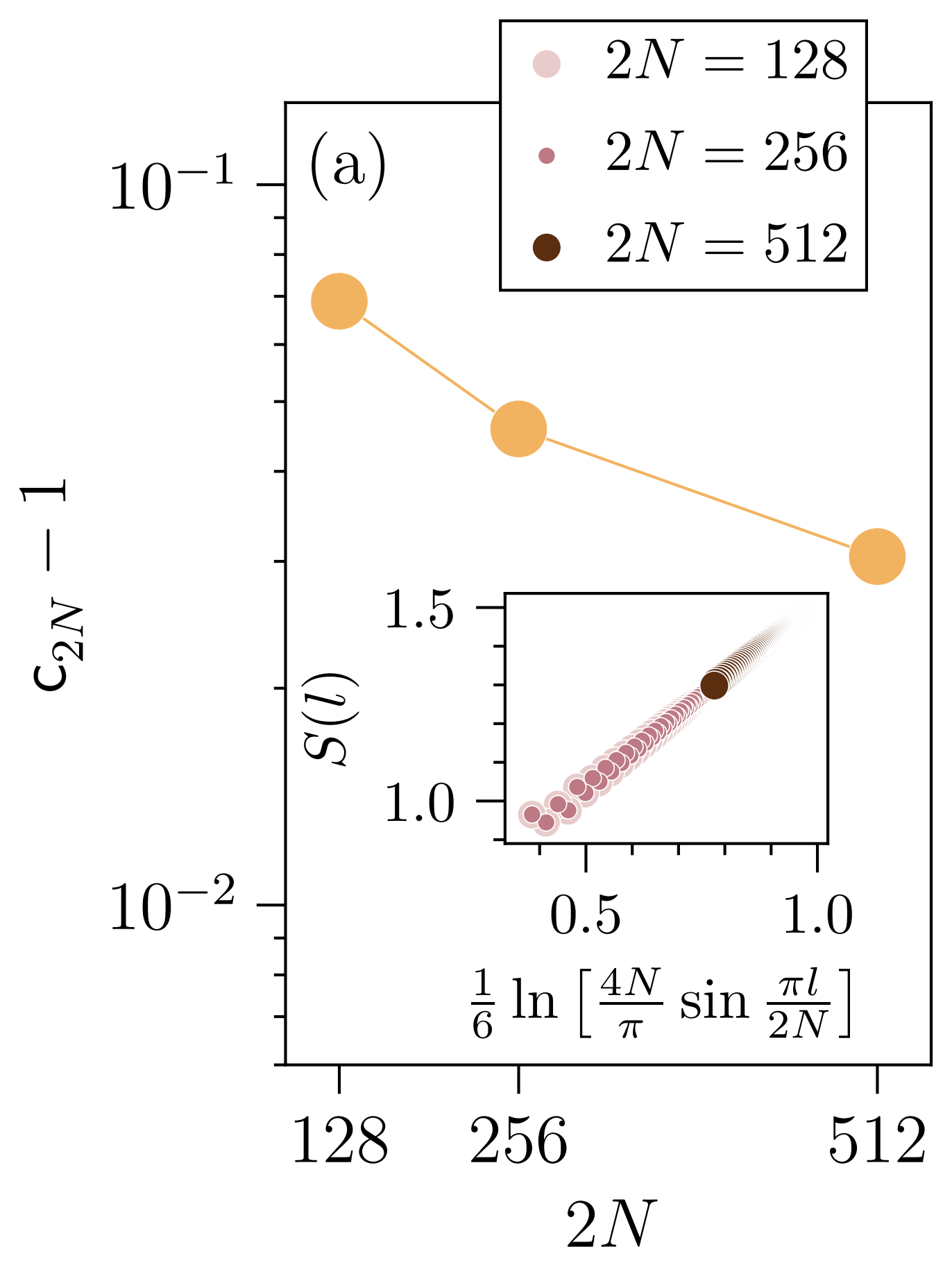

(a)
$2N = 128$
$2N = 256$
$2N = 512$
10^{-1}
10^{-2}
$c_{2N} - 1$
128
256
512
$2N$
1.5
1.0
$S(l)$
0.5
1.0
$\frac{1}{6} \ln \left[\frac{4N}{\pi} \sin \frac{\pi l}{2N} \right]$